ADVANCED ATTITUDE DETERMINATION AND CONTROL SYSTEM (ADCS)

With Extended Kalman Filter and Failure Mode Analysis

Report submitted by

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Abstract

This project presents the modeling and simulation of an Advanced Attitude Determination and Control System (ADCS) for satellite applications. The system integrates quaternion-based rigid body dynamics, a Reaction Wheel actuator model with torque saturation, Proportional-Derivative (PD) attitude control, and an Extended Kalman

Filter (EKF) for real-time attitude and gyro bias estimation. A star tracker is incorporated as a high-accuracy attitude sensor, and a failure scenario is simulated after 40 seconds to evaluate system robustness. Random disturbance torques are introduced to emulate realistic space environmental effects.

Introduction

Modern satellites require precise attitude control for communication, Earth observation, remote sensing, and scientific missions. This project develops a complete ADCS simulation incorporating actuator saturation, disturbance torques, gyro bias estimation, and sensor failure handling.

Literature Survey

Extensive research has been conducted in spacecraft attitude estimation and control.

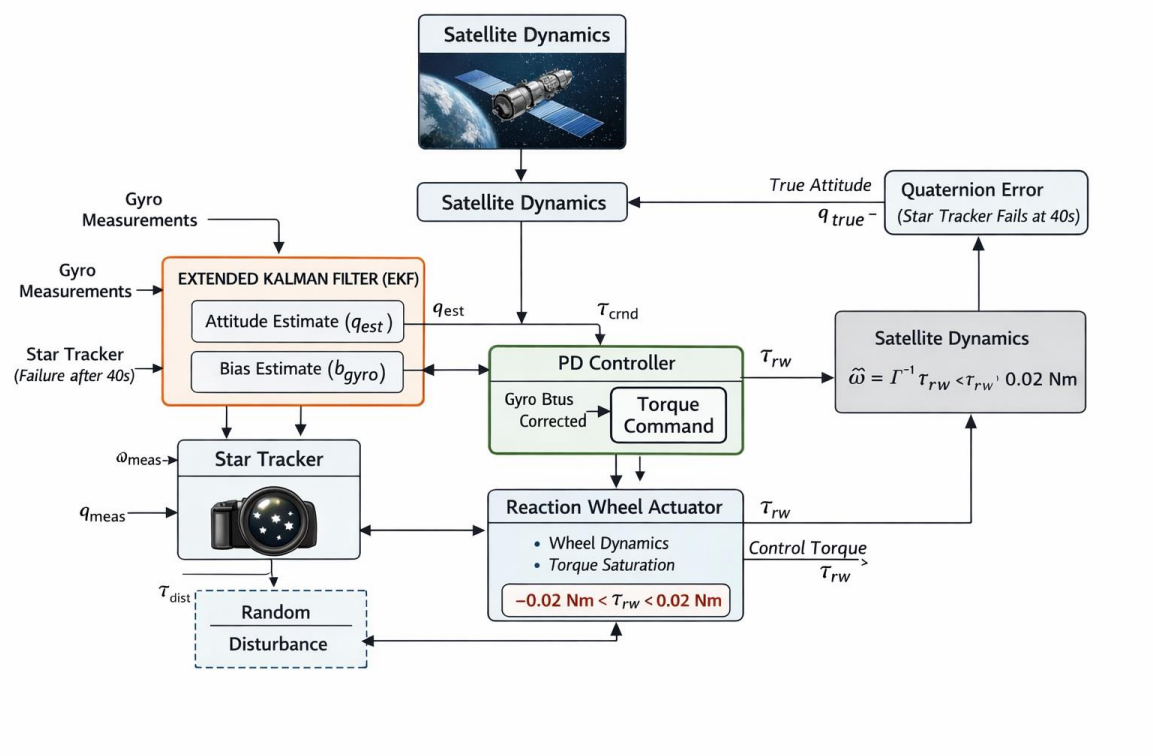
- Wertz (1978) introduced classical spacecraft attitude determination principles.
- Shuster (1981) developed quaternion-based solutions for Wahba's problem.
- Extended Kalman Filter (EKF) has been widely used in satellite missions for nonlinear state estimation.
- Reaction wheel-based control systems are standard in small satellite missions due to precision and reliability.
- Recent studies focus on fault-tolerant ADCS and safe-mode operation during sensor failures.

  However, many academic simulations do not incorporate combined actuator saturation, disturbance modelling , and failure mode analysis simultaneously. This project addresses that gap through integrated modelling.

Proposed System

The proposed ADCS integrates satellite rigid body dynamics, quaternion representation, sensor fusion using EKF, PD control, reaction wheel actuator with saturation, disturbance modeling, and safe-mode operation after star tracker failure.

System Flowchart



Methodology

1. Initialize spacecraft states.
2. Propagate dynamics.
3. Simulate sensor measurements.
4. Perform EKF prediction and correction.
5. Apply PD control.
6. Enforce actuator saturation.
7. Inject disturbance torque.
8. Switch to safe mode after sensor failure.

Results

The simulation results demonstrate the performance of the **satellite Attitude Determination and Control System (ADCS)** using the **Extended Kalman Filter (EKF)** and **Reaction Wheel control**. Several parameters were analyzed to evaluate the stability and accuracy of the system.

Attitude Error

The attitude error graph shows the difference between the **desired orientation** and the **estimated satellite orientation**. Initially, the error is higher due to system initialization and disturbances. As the EKF correction and control system operate, the error gradually decreases and stabilizes near zero, indicating accurate attitude estimation and control.

Quaternion Response

The quaternion values represent the satellite orientation in space. The simulation shows that the quaternion states converge smoothly over time, proving that the system can maintain a stable orientation without singularity issues.

Reaction Wheel Torque

The reaction wheel torque graph represents the control effort applied to stabilize the satellite. The torque initially fluctuates to correct the orientation but later becomes stable once the satellite reaches the desired attitude.

EKF Estimation Error

The EKF estimation error illustrates how well the filter estimates the true attitude. The error decreases as more

measurements from the sensors (such as gyroscope and star tracker) are processed, showing that the EKF effectively reduces noise and improves estimation accuracy.

Future Enhancement

Future improvements include implementing Unscented Kalman Filter (UKF), adding magnetorquer-based momentum dumping, modeling environmental torques such as gravity gradient and solar radiation pressure, integrating Fault Detection and Isolation (FDI), and performing hardware-in-the-loop validation.

Conclusion

The developed ADCS demonstrates stable attitude control, accurate bias estimation, and robust failure handling, reflecting practical small satellite ADCS architecture.